

Equivalent Dichotomies for Triangle Detection in Subgraph, Induced, and Colored H -Free Graphs

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Triangle Detection

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Given an n -vertex graph G , does G contain a triangle?



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No other truly subcubic $n^{3-\Omega(1)}$ -time algorithm is known!

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- All-Pairs Shortest Paths in $n^{3-\Omega(1)}$ time?

APSP $\Rightarrow$ **Minimum-Weight Triangle**

- 3SUM in $n^{2-\Omega(1)}$ time?

3SUM $\Rightarrow$ **Listing Triangles**

- Online Matrix-Vector Multiplication in $n^{3-\Omega(1)}$ time?

OMv $\Rightarrow$ **Online Triangle Detection**

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FMM does not address **weighted**, **listing**, and **online** variants of Triangle Detection.

Triangle Detection in H -Free Graphs

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Example: A complete bipartite graph is triangle-free.

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For which patterns H can Triangle Detection in H -free graphs be solved by a combinatorial algorithm in $n^{3-\Omega(1)}$ time?

The Classification Objective

The Objective

For each pattern H , determine which of the following holds:

1. Triangle Detection in H -free graphs is solvable in $n^{3-\Omega(1)}$ time.
2. Triangle Detection in H -free graphs is as hard as Triangle Detection in general graphs.

Our previous results [ITCS 2026]:

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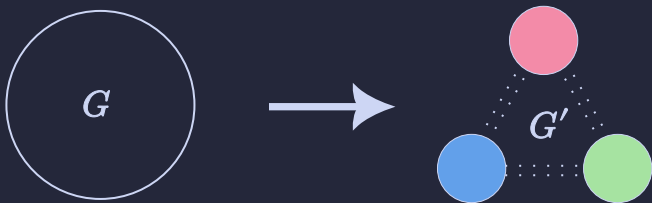
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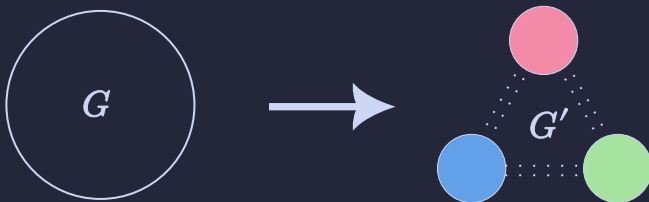


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Color the vertices randomly with 3 colors; delete monochromatic edges.

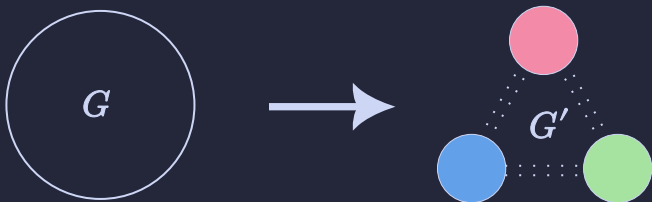
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- If G contains a triangle, it survives with probability $\Omega(1)$.
- Repeating this $O(\log n)$ times succeeds with high probability.
- Every produced subgraph is H -free for every pattern H with $\chi(H) > 3$.

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For $\chi(H) > 3$, Triangle Detection in H -free graphs remains as hard as Triangle Detection in general graphs.

Lower Bounds: The Case of Two or More Triangles in H

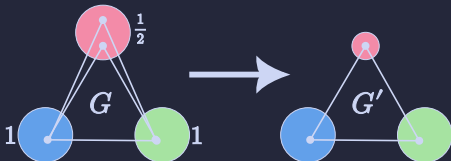
Idea: Well-known self-reduction to Unique Triangle Detection.



Random sieve: for each $i, j, k \in [\log_2 n]$, retain each red vertex with probability 2^{-i} , each blue vertex with probability 2^{-j} , and each green vertex with probability 2^{-k} .

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Random sieve: for each $i, j, k \in [\log_2 n]$, retain each **red** vertex with probability 2^{-i} , each **blue** vertex with probability 2^{-j} , and each **green** vertex with probability 2^{-k} .

- The number of produced subgraphs is $\text{poly}(\log n)$.
- (NO-case) If G is triangle-free, then every produced graph is triangle-free and therefore H -free.
- (YES-case) If G contains a triangle, some produced graph G' has a unique triangle and therefore G' is H -free.

The Dichotomy Hypothesis

In our ITCS 2026 paper, we propose the following hypothesis:

The Dichotomy Hypothesis

Triangle Detection in H -free graphs is classified as follows:

- **Hard** if H is not 3-colorable or contains at least two triangles; and
- **Easy**, otherwise.

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Results So Far

- For every pattern in the **hard** class, the problem has been confirmed to remain as hard as Triangle Detection in general graphs.
- An $n^{3-\Omega(1)}$ -time algorithm is known only for a **subset** of the patterns in the **easy** class.

This Paper: Other Notions of Structure?

This investigation is also well-motivated in the induced setting.

Definition

G is induced H -free if it contains no induced subgraph isomorphic to H .

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Note: Induced H -free graphs are typically more complex.

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The following inclusions are straightforward:

$$\begin{array}{ccc} H\text{-free graphs} & \subseteq & \text{induced } H\text{-free graphs} \\ & \Downarrow & \\ \text{Easy}(\text{induced}) & \subseteq & \text{Easy}(\text{subgraph}) \\ \text{Hard}(\text{subgraph}) & \subseteq & \text{Hard}(\text{induced}) \end{array}$$

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A Priori Possibility

There could be a pattern H for which H -free graphs are **easy**, but induced H -free graphs are **hard**.

$$H \in \text{Easy}(\text{subgraph}) \cap \text{Hard}(\text{induced})$$

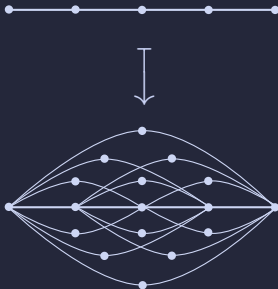
Induced-to-Subgraph Reduction

Theorem

There is a mapping $H \mapsto H^+$ preserving the “easy” properties:

H is 3-colorable $\implies H^+$ is 3-colorable

H has at most one triangle $\implies H^+$ has at most one triangle

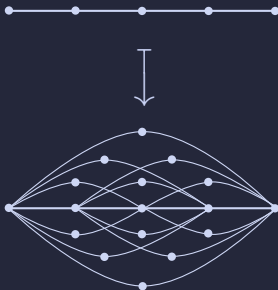


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and an $n^{3-\Omega(1)}$ -time reduction from Triangle Detection in induced H -free graphs to Triangle Detection in (subgraph) H^+ -free graphs.

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In particular, assuming that the dichotomy hypothesis holds in the subgraph setting, we have:

$$\text{Easy}(\text{subgraph}) = \text{Easy}(\text{induced})$$

$$\text{Hard}(\text{subgraph}) = \text{Hard}(\text{induced})$$

Takeaway: it is enough to focus on the (simpler) subgraph setting.

Example: From Induced P_5 -Free to P_5^+ -Free

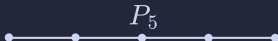
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Example: From Induced P_5 -Free to P_5^+ -Free

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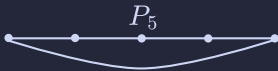
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3. Otherwise, G contains a copy of P_5 with at least one extra edge.



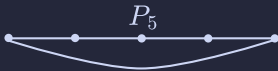
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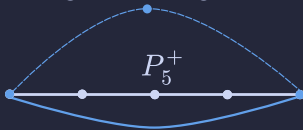


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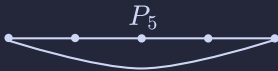
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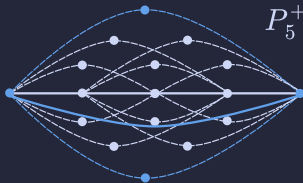
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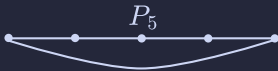
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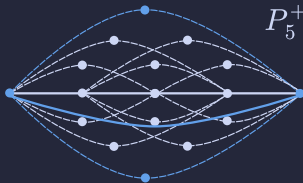
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P_5^+ in $G \implies G$ contains at least two triangles.

Intuitively, the hardest instances have at most one triangle.

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Theorem

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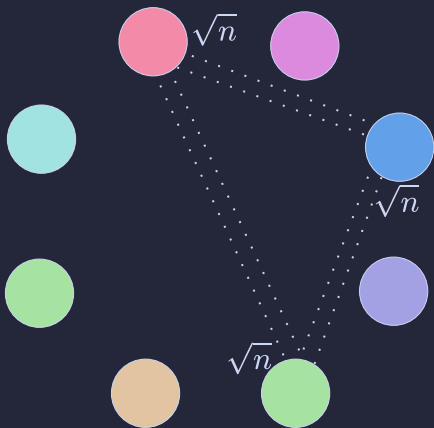


The Color-Coding Challenge

Randomly partitioning vertices and deleting monochromatic edges does **not preserve** induced H -freeness.

Color-Coding That Preserves Induced H -Freeness

Algorithmic Ramsey: Find either a triangle or an $O(\sqrt{n})$ -coloring of G in linear time. Split the color classes into parts of size $O(\sqrt{n})$.



Every triple of color classes **induces** a 3-colored subgraph of G , and at least one such subgraph contains a triangle, if one exists.

Potential Follow-Ups

Open Questions:

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in Subgraph, Induced, and Colored H -Free
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